

# 13 - Intro to organic chemistry

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Organic chemistry is the study of organic compounds

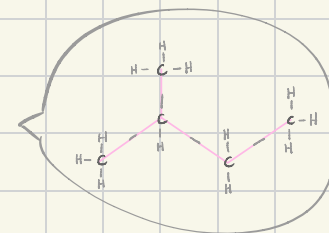
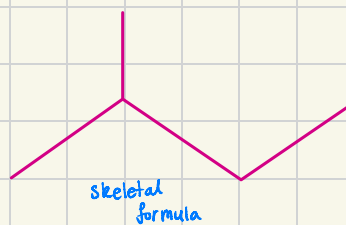
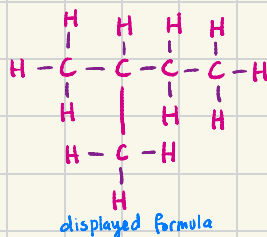
→ compounds that contain carbon (& hydrogen)

## Formulae for organic compounds

- i) structural formula → shows the order of the atoms joined together in a compound
- ii) displayed formula → shows the order of the atoms joined together & their orientation/bond angles
- iii) skeletal formula → simplifies carbon & hydrogen & only shows functional groups

e.g 2-methylpentane

structural formula  
 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$



## Functional groups

\* a functional group is a group of atoms in an organic compound that determine the chemical properties of the compound

→ a homologous series is a series of compounds with the same functional group & they differ in the number of carbon atoms

| Homologous series | Functional group | prefix/suffix                                  | structural formula                                                           |
|-------------------|------------------|------------------------------------------------|------------------------------------------------------------------------------|
| alkene            | C=C bond         | -ene                                           | $\text{R}-\text{C}=\text{C}-\text{R}$<br><small>rest of the compound</small> |
| halogenoalkane    | halogen          | halo-<br><small>depends on the halogen</small> | $\text{R}-\text{X}$<br><small>any halogen</small>                            |
| alcohol           | hydroxyl         | -ol                                            | $\text{R}-\text{OH}$                                                         |
| aldehyde          | carbonyl         | -al                                            | $\text{R}-\text{C}(=\text{O})\text{H}$                                       |
| ketone            | carbonyl         | -one                                           | $\text{R}-\text{C}(=\text{O})-\text{R}'$                                     |
| carboxylic acid   | carboxyl         | -oic acid                                      | $\text{R}-\text{C}(=\text{O})\text{OH}$                                      |
| ester             | ester            | -oate                                          | $\text{R}-\text{C}(=\text{O})\text{OR}$                                      |

| eg. | name                               | displayed formula                                                                                         | skeletal formula |
|-----|------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------|
|     | propene                            | $\text{H}-\text{C}(\text{H})=\text{C}(\text{H})-\text{H}$                                                 |                  |
|     | if X → chlorine<br>1-chloropropane | $\text{H}-\text{C}(\text{H})-\text{C}(\text{H})-\text{C}(\text{H})-\text{X}$                              |                  |
|     | propan-1-ol                        | $\text{H}-\text{C}(\text{H})-\text{C}(\text{H})-\text{C}(\text{H})-\text{OH}$                             |                  |
|     | propanal                           | $\text{H}-\text{C}(\text{H})-\text{C}(\text{H})-\text{C}(=\text{O})\text{H}$                              |                  |
|     | propanone                          | $\text{H}-\text{C}(\text{H})-\text{C}(=\text{O})-\text{C}(\text{H})-\text{H}$                             |                  |
|     | propanoic acid                     | $\text{H}-\text{C}(\text{H})-\text{C}(\text{H})-\text{C}(=\text{O})\text{OH}$                             |                  |
|     | methyl propanoate                  | $\text{H}-\text{C}(\text{H})-\text{O}-\text{C}(=\text{O})-\text{C}(\text{H})-\text{C}(\text{H})-\text{H}$ |                  |

|         |         |          |                |
|---------|---------|----------|----------------|
| amine   | amine   | -amine   | $R-NH_2$       |
| nitrile | nitrile | -nitrile | $R-C \equiv N$ |

|                |                                                                                              |  |
|----------------|----------------------------------------------------------------------------------------------|--|
| propylamine    | $\begin{array}{c} H & H & H \\   &   &   \\ H-C-C-C-N \\   &   &   \\ H & H & H \end{array}$ |  |
| propanenitrile | $\begin{array}{c} H & H \\   &   \\ H-C-C-C \equiv N \\   &   \\ H & H \end{array}$          |  |

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## Hydrocarbons

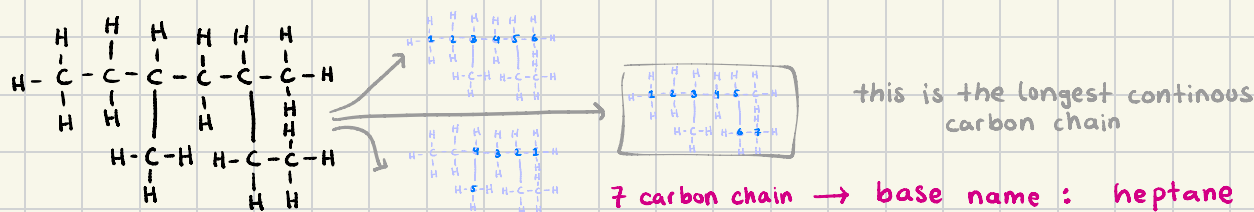
→ compounds made up of carbon & hydrogen atoms

→ alkanes are hydrocarbons with no functional group

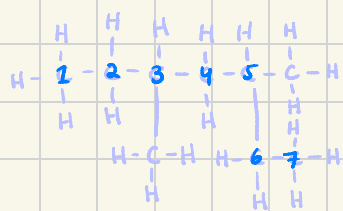
## Naming compounds

• compounds are named using systematic nomenclature (IUPAC naming)

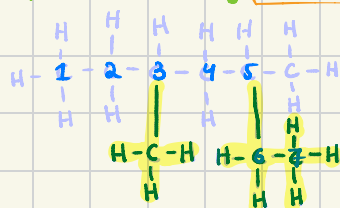
Step 1: identify the longest carbon chain (parent chain) - determines the base name of compound



Step 2: number the parent chain



Step 3: identify **substituents** ← groups of atoms that branch off of the main carbon chain



at carbon #3 there is a methyl group ( $-CH_3$ )

at carbon #5 there is an ethyl group ( $-CH_2CH_3$ )

Step 3: name the compound

parent chain: heptane

substituents: methyl at carbon #3 & ethyl at carbon #5] substituents are arranged alphabetically  
→ ethyl and then methyl

Name: - **5-ethyl-3-methylheptane**

## Saturated compounds

compounds that only have single bonds

## Unsaturated compounds

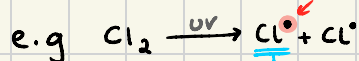
compounds that contain double/triple bonds

## Fission

- a process through which chemical bonds are broken, resulting in the formation of two or more species
- two kinds: homolytic and heterolytic

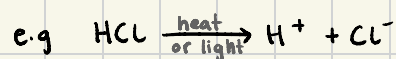
covalent bond breaks evenly & each atom involved in the bond retains one of the two electrons that were previously shared in the bond

represents a free radical



free radical: species with an unpaired electron

covalent bond breaks unevenly & one of the atoms takes both of the electrons that were originally shared in the bond. This results in a formation of a cation & anion



### ★ steps of free radical reaction:

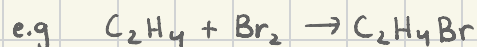
- 1) initiation: formation of the free radicals - energy is required to break the covalent bonds
- 2) propagation: radicals attack the reactant molecules to generate more free radicals
- 3) termination: two radicals meet & form a product molecule, leading to the end of reaction

free radical substitution: reaction where a free radical replaces an atom/group in a molecule  
\* often seen in the halogenation alkanes

## Types of organic reactions

### i) addition reactions

formation of a single product from two or more reactants



### ii) elimination reactions

removal of a small molecule from a larger reactant molecule

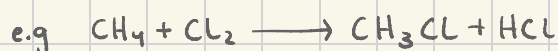


### iii) condensation reactions

reactions where addition & then elimination take place e.g.  $\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$

### iv) substitution reactions

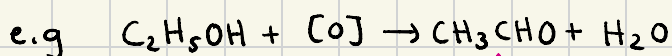
involve the replacement of an atom(s) by another



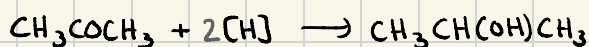
Oxidation: loss of electrons or increase in oxidation state, involving the addition of oxygen or removal of hydrogen

Reduction: gain of electrons or decrease in oxidation state, involving the removal of oxygen or addition of hydrogen

in organic redox reactions, [O] represents an atom of oxygen from an oxidising agent & [H] represents an atom of hydrogen from a reducing agent



ethanol loses two hydrogens & has thus been oxidised



\* don't forget to balance equation

## Nucleophiles

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- an electron-rich species that donates electrons

nucleophilic reactions: reactions where nucleophiles donate electrons

nucleophilic substitution: reaction where a nucleophile replaces a leaving group in a molecule

nucleophilic addition: reaction where a nucleophile adds to a molecule with polar bonds

both result in the formation of a new covalent bond

## Electrophiles

- an electron-deficient species that accepts electrons

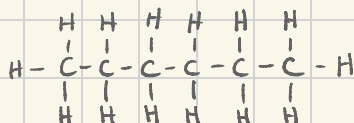
electrophilic reactions: reactions where electrophiles accept electrons

electrophilic addition: reaction where an electrophile adds to a molecule with double/triple bonds

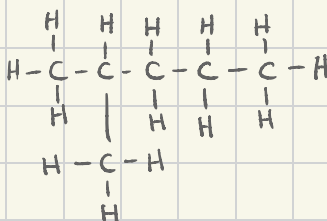
## Shapes of organic molecules

- can either be

straight-chained

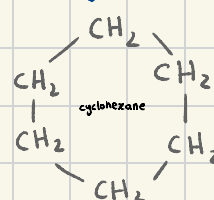


branched



or

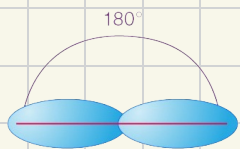
cyclic



- in hybridised atoms

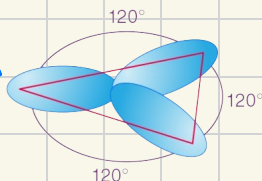
$sp$

linear shape with a  $180^\circ$  bond angle



$sp^2$

trigonal planar shape with a  $120^\circ$  bond angle



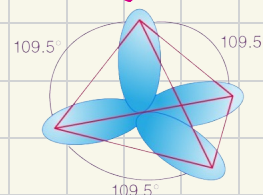
$sp$  &  $sp^2$  atoms are all planar

made of:

- 1 sigma ( $\sigma$ ) bond
- 2 pi ( $\pi$ ) bonds

$sp^3$

tetrahedral shape with a bond angle of  $109.5^\circ$



made of:

only sigma ( $\sigma$ ) bonds



## Isomers:

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- two or more compounds with the same molecular formula but a different arrangement of atoms or their spatial orientation. Organic molecules that exhibit this property show isomerism
- two types: structural isomerism and stereoisomerism

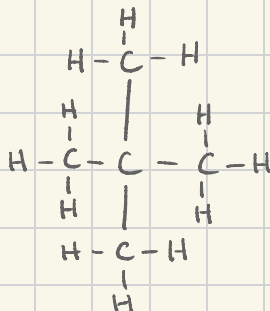
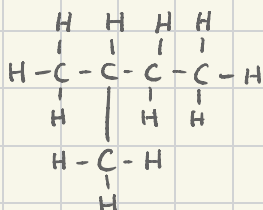
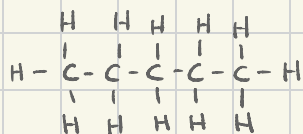
### Structural isomerism

\* compounds with the same molecular formula but a different arrangement of atoms

three types

#### 1) chain isomerism

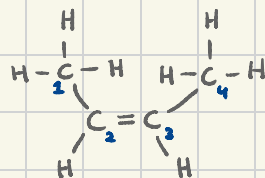
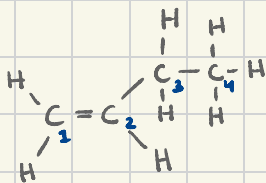
↳ different arrangements of the carbon skeleton



all 3 have the  
molecular formula:  $C_5H_{12}$

#### 2) positional isomerism

↳ variation in position of functional group/substituent



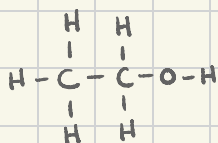
double bond in 1-2

double bond in 2-3

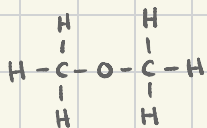
both have the  
molecular formula:  $C_4H_8$

#### 3) functional isomerism

↳ different functional groups within compound



alcohol



ether  
(don't need to know)

both have the  
molecular formula:  $C_2H_6O$

## Stereoisomerism

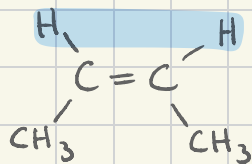
\* compounds with the same molecular formula but a different arrangement of their spatial orientation

### two types

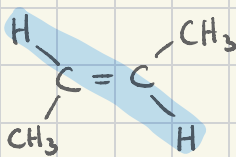
#### 1) geometrical isomerism

↳ occurs in molecules with restricted rotation (due to a double bond; presence of a  $\pi$  bond)

↳ two kinds: cis (substituents on the same side)  $\neq$  trans (substituents on opposite sides)



cis-2-buten

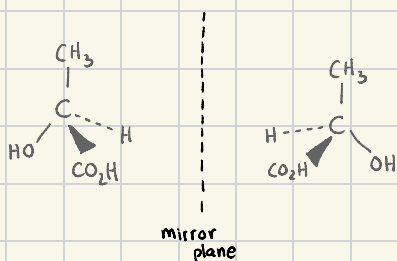


trans-2-buten

#### 2) optical isomerism

↳ occurs when two molecules are mirror images of each other but cannot be superimposed

↳ exist because of chiral centres in a molecule



these are optical isomers, referred to as enantiomers.

a chiral centre refers to a carbon atom bonded to four different groups

chiral centres give rise to enantiomers

\* compounds can contain more than one chiral centre